Lctr 19: The z-Transform: Definition and Region of Convergence

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Spring 2026

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Focus: Sections 10.0–10.2 (Pages 741–757)

19.1 The Big Picture: The DT Counterpart of the Laplace Transform

Why This Matters

In Lectures 16–18 we developed the Laplace transform for continuous-time signals. It generalized the CT Fourier transform by replacing $j\omega$ with the complex variable $s = \sigma + j\omega$, letting us analyze signals that the Fourier transform cannot handle (growing exponentials, unstable systems).

We now need the **exact same generalization for discrete time**. The DTFT (from Lecture 12) requires $\sum |x[n]| < \infty$, so a growing sequence like $x[n] = 2^n\, u[n]$ has no DTFT. The **z-transform** fixes this by introducing a complex variable z that plays the same role in DT that s plays in CT. The parallel is systematic:

Laplace (CT)	**z-Transform (DT)**
Complex variable $s = \sigma + j\omega$	Complex variable $z = re^{j\omega}$
$j\omega$-axis ($\sigma = 0$)	Unit circle ($\lvert z\rvert = 1$)
Left/Right half-plane	Inside/outside the unit circle
ROC: vertical strip	ROC: annular ring
Differential equations $\to$ algebra	Difference equations $\to$ algebra

Because you already understand the Laplace transform, learning the z-transform is substantially faster. This lecture focuses on what is *different* between the two.

Roadmap:

1. Motivate and define the z-transform.
2. Explain the z-plane and the role of the unit circle.
3. Compute z-transforms and ROCs for fundamental DT signals.
4. State the ROC properties.
5. Summarize the complete CT/DT parallel.

19.2 From the DTFT to the z-Transform

19.2.1 The Convergence Problem

Recall the DTFT from Lecture 12:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]\, e^{-j\omega n}$$

This requires **absolute convergence**: $\sum_{n=-\infty}^{\infty} |x[n]| < \infty$. Many important sequences violate this condition. For instance, $x[n] = 2^n u[n]$ gives $\sum_{n=0}^{\infty} 2^n = \infty$.

19.2.2 The Fix: Multiply by a Geometric Damping Factor

In the Laplace transform, we multiplied $x(t)$ by $e^{-\sigma t}$ to force convergence. The DT analog is to multiply $x[n]$ by r^{-n}, where $r > 0$ is a real constant. Apply the DTFT to the damped sequence $x[n]\, r^{-n}$:

$$\sum_{n=-\infty}^{+\infty} \left[x[n]\, r^{-n}\right] e^{-j\omega n} = \sum_{n=-\infty}^{+\infty} x[n] \left(r\, e^{j\omega}\right)^{-n}$$

If we choose r large enough, $x[n]\, r^{-n}$ may decay fast enough for the sum to converge. Now define the complex variable:

$$\boxed{z = r\, e^{j\omega}}$$

where $r = |z|$ is the magnitude and $\omega = \angle z$ is the angle. Substituting gives the z-transform:

$$\boxed{X(z) = \sum_{n=-\infty}^{+\infty} x[n]\, z^{-n}}$$

Key Insight

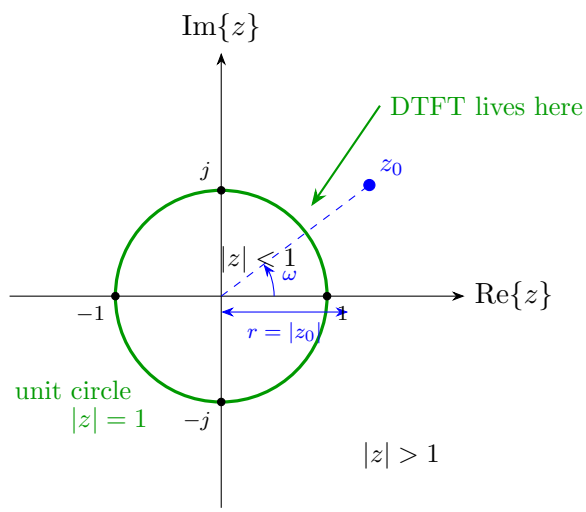
The z-transform is the DTFT of $x[n]\, r^{-n}$. When $r = 1$ (i.e., $|z| = 1$), the damping disappears and the z-transform reduces exactly to the DTFT:

$$X(z)\big|_{z=e^{j\omega}} = X(e^{j\omega}) \qquad \text{(the DTFT)}$$

The **unit circle** in the z-plane plays the same role as the $j\omega$-axis in the s-plane: it is where the Fourier transform lives.

19.2.3 The z-Plane

Since $z = re^{j\omega}$ is complex, we visualize it on the z-plane:



Three facts to remember about the z-plane:

1. Points on the unit circle have $|z| = 1$, so $z = e^{j\omega}$. This is where the DTFT lives.
2. $z = 1$ corresponds to $\omega = 0$ (DC). $z = -1$ corresponds to $\omega = \pi$ (Nyquist frequency).
3. Points inside the unit circle have $|z| < 1$; points outside have $|z| > 1$.

19.3 Formal Definition

The z-Transform Pair

Analysis (forward transform):

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n]\, z^{-n}$$

Synthesis (inverse transform):

$$x[n] = \frac{1}{2\pi j} \oint X(z)\, z^{n-1}\, dz$$

where the contour integral is along a closed path in the ROC that encircles the origin.
Notation: $x[n] \;\mathcal{Z}\; X(z)$

Warning

Just as with the Laplace transform, $X(z)$ alone does **not** uniquely determine $x[n]$. The Region of Convergence (ROC), the set of z values where the sum converges absolutely, must also be specified. Different signals can share the same $X(z)$ but have different ROCs.

19.4 Fundamental z-Transform Pairs

19.4.1 Right-Sided Exponential: $x[n] = a^n\, u[n]$

Right-Sided Geometric Sequence

Signal: $x[n] = a^n\, u[n]$.
Step 1: Write the sum. Since $u[n] = 0$ for $n < 0$:

$$X(z) = \sum_{n=0}^{\infty} a^n\, z^{-n} = \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n = \sum_{n=0}^{\infty} (az^{-1})^n$$

Step 2: Recognize the geometric series. The sum $\sum_{n=0}^{\infty} w^n = \frac{1}{1-w}$ converges if and only if $|w| < 1$. Here $w = az^{-1} = a/z$, so we need:

$$|az^{-1}| < 1 \quad \Longleftrightarrow \quad |z| > |a|$$

When this condition holds:

$$X(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$

Step 3: Identify pole, zero, and ROC.
Pole at $z = a$, zero at $z = 0$. ROC: $|z| > |a|$ (exterior of a circle of radius $|a|$).

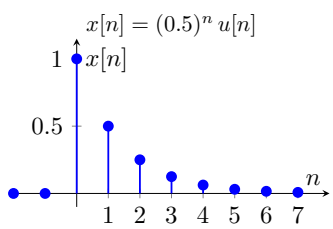
$$\boxed{a^n\, u[n] \;\; \mathcal{Z} \;\; \frac{1}{1 - az^{-1}} = \frac{z}{z - a}\,, \qquad |z| > |a|}$$

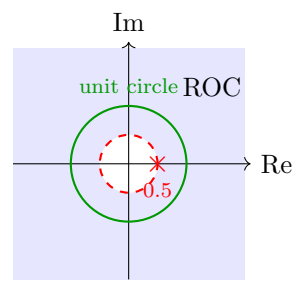
Numerical: Decaying Sequence a

With $a = 0.5$:

$$(0.5)^n\, u[n] \;\; \mathcal{Z} \;\; \frac{1}{1 - 0.5\, z^{-1}}\,, \qquad |z| > 0.5$$

Pole at $z = 0.5$. The unit circle ($|z| = 1 > 0.5$) is in the ROC, so the DTFT exists.
Sanity checks:
At $z = 1$ (DC): $X(1) = \frac{1}{1-0.5} = 2$. Direct sum: $\sum_{n=0}^{\infty}(0.5)^n = \frac{1}{1-0.5} = 2$. ✓
At $z = -1$ (Nyquist): $X(-1) = \frac{1}{1+0.5} = \frac{2}{3}$. Direct sum: $\sum_{n=0}^{\infty}(-0.5)^n = \frac{1}{1.5} = \frac{2}{3}$. ✓





Numerical: Growing Sequence a

With $a = 2$:

$$2^n\, u[n] \;\; \mathcal{Z} \;\; \frac{1}{1 - 2z^{-1}}\,, \qquad |z| > 2$$

Pole at $z = 2$. The unit circle ($|z| = 1$) is *not* in the ROC ($|z| > 2$), so the DTFT does not exist. This makes sense: $\sum_{n=0}^{\infty} 2^n$ diverges, so the sequence is not absolutely summable. This is the power of the z-transform: it can handle sequences whose DTFT does not exist, just as the Laplace transform handles growing exponentials.

19.4.2 Left-Sided Exponential: $x[n] = -a^n\, u[-n-1]$

Left-Sided Geometric Sequence

Signal: $x[n] = -a^n\, u[-n-1]$. This equals $-a^n$ for $n \le -1$ and 0 for $n \ge 0$.
Step 1: Write the sum.

$$X(z) = \sum_{n=-\infty}^{-1} (-a^n)\, z^{-n} = -\sum_{n=-\infty}^{-1} (az^{-1})^n$$

Step 2: Substitute $m = -n$ so that m runs from 1 to ∞:

$$X(z) = -\sum_{m=1}^{\infty} (a^{-1}z)^m = -\sum_{m=1}^{\infty} \left(\frac{z}{a}\right)^m$$

This is a geometric series in $w = z/a$, starting at $m = 1$. It converges iff $|z/a| < 1$, i.e., $|z| < |a|$.
Step 3: Evaluate. Using $\sum_{m=1}^{\infty} w^m = \frac{w}{1-w}$ for $|w| < 1$:

$$X(z) = -\frac{z/a}{1 - z/a} = -\frac{z}{a-z} = \frac{z}{z-a} = \frac{1}{1-az^{-1}}$$

$$\boxed{-a^n\, u[-n-1] \;\; \mathcal{Z} \;\; \frac{1}{1-az^{-1}}\,, \qquad |z| < |a|}$$

Warning

The right-sided pair $a^n u[n]$ and the left-sided pair $-a^n u[-n-1]$ produce the **same** algebraic expression $\frac{1}{1-az^{-1}}$. The ROC is the *only* difference:

Signal	$X(z)$	**ROC**
$a^n\, u[n]$	$\dfrac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$ (exterior)
$-a^n\, u[-n-1]$	$\dfrac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$ (interior)

Without the ROC, the transform is ambiguous. **Always state the ROC.**

Numerical: Left-Sided with a

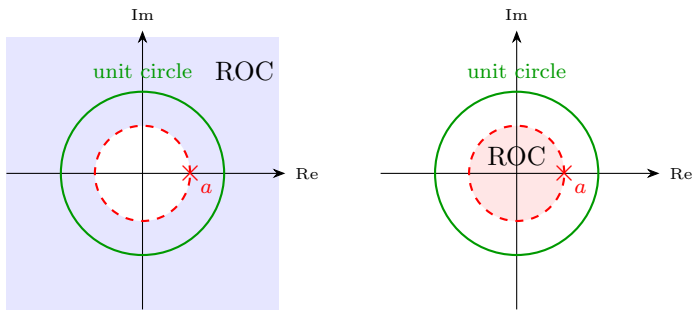
$x[n] = -(3)^n\, u[-n-1]$. With $a = 3$:

$$X(z) = \frac{1}{1-3z^{-1}}\,, \qquad |z| < 3$$

The signal values are: $x[-1] = -3^{-1} = -1/3$, $x[-2] = -3^{-2} = -1/9$, $x[-3] = -1/27, \ldots$
The sequence decays as $n \to -\infty$, which is why it is absolutely summable.
The pole is at $z = 3$ (outside the unit circle). The ROC ($|z| < 3$) *includes* the unit circle, so

the DTFT exists. ✓

19.4.3 ROC Diagrams: Right-Sided vs. Left-Sided



Right-sided: $|z| > |a|$ **Left-sided:** $|z| < |a|$

19.4.4 The Unit Impulse and Unit Step

Unit Impulse and Delayed Impulse

Unit impulse: $\delta[n] \;\; \mathcal{Z} \;\; 1$, ROC: all z.
The only nonzero term in the sum is $n = 0$, giving $\delta[0] \cdot z^0 = 1$.
Delayed impulse: $\delta[n-k] \;\; \mathcal{Z} \;\; z^{-k}$, ROC: $|z| > 0$ (for $k \geq 1$).
Each unit of delay contributes a factor of z^{-1}. A delay of k samples gives z^{-k}, which has k zeros at $z = 0$ and a k-th order pole at $z = 0$ when viewed as $1/z^k$.

Unit Step

The unit step $u[n]$ is the special case $a^n u[n]$ with $a = 1$:

$$u[n] \;\; \mathcal{Z} \;\; \frac{1}{1 - z^{-1}} = \frac{z}{z-1}, \qquad |z| > 1$$

The pole is at $z = 1$, which sits exactly on the unit circle. The ROC does not include the unit circle (the boundary $|z| = 1$ is excluded by the strict inequality $|z| > 1$), so the DTFT of $u[n]$ cannot be obtained by simply setting $z = e^{j\omega}$ in the z-transform. (Recall from Lecture 12 that the DTFT of $u[n]$ requires a special treatment with impulse terms.)

19.4.5 Two-Sided Signals

Two-Sided Signal with Annular ROC

Signal: $x[n] = (0.5)^n\, u[n] + 2 \cdot (3)^n\, u[-n-1]$.
Step 1: Transform each piece separately.
Right-sided part:

$$(0.5)^n u[n] \;\; \mathcal{Z} \;\; \frac{1}{1-0.5z^{-1}}\,, \quad |z| > 0.5$$

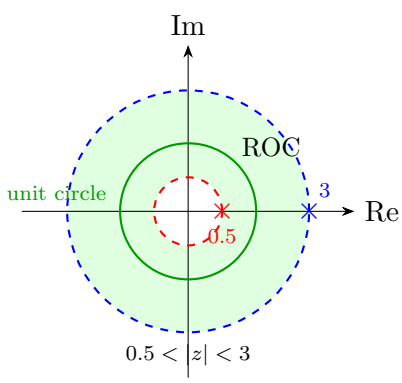
Left-sided part: $2 \cdot (3)^n u[-n-1] = -2 \cdot \big[-(3)^n u[-n-1]\big]$, so:

$$\mathcal{Z} \;\; \frac{-2}{1-3z^{-1}}\,, \quad |z| < 3$$

Step 2: Add and intersect the ROCs.

$$X(z) = \frac{1}{1-0.5z^{-1}} - \frac{2}{1-3z^{-1}}\,, \qquad 0.5 < |z| < 3$$

The ROC is the intersection $\{|z| > 0.5\} \cap \{|z| < 3\}$, which is the annular ring $0.5 < |z| < 3$. Since $0.5 < 1 < 3$, the unit circle lies inside this ring, so the DTFT exists. ✓



Key Insight

The ROC for a two-sided signal is a ring $r_1 < |z| < r_2$ between two pole circles. If the inner pole has a larger radius than the outer pole (e.g., requiring $|z| > 5$ and $|z| < 2$ simultaneously), the intersection is empty and the z-transform does not exist.

19.4.6 Sum of Two Right-Sided Exponentials

Combining Two Right-Sided Terms

Signal: $x[n] = 3(0.4)^n u[n] - 2(0.8)^n u[n]$.

Step 1: Transform each term:

$$\frac{3}{1-0.4z^{-1}}\,,\ |z|>0.4 \qquad \text{and} \qquad \frac{-2}{1-0.8z^{-1}}\,,\ |z|>0.8$$

Step 2: Common denominator:

$$X(z)=\frac{3(1-0.8z^{-1})-2(1-0.4z^{-1})}{(1-0.4z^{-1})(1-0.8z^{-1})}=\frac{1-1.6z^{-1}}{(1-0.4z^{-1})(1-0.8z^{-1})}$$

Step 3: ROC is the intersection $|z|>0.4\ \cap\ |z|>0.8=|z|>0.8$.
Poles: $z=0.4$ and $z=0.8$. Zero: from $1-1.6z^{-1}=0$, so $z=1.6$.
Check: $X(1)=\frac{1-1.6}{(1-0.4)(1-0.8)}=\frac{-0.6}{(0.6)(0.2)}=-5$. Direct: $\frac{3}{0.6}-\frac{2}{0.2}=5-10=-5$. ✓

19.4.7 Finite-Duration Sequence

Polynomial in z^{-1}

Signal: $x[n]=\{2,\ -1,\ 3,\ 0,\ 4\}$ for $n=0,1,2,3,4$ (zero elsewhere).

$$X(z)=2\cdot z^0+(-1)\cdot z^{-1}+3\cdot z^{-2}+0\cdot z^{-3}+4\cdot z^{-4}=2-z^{-1}+3z^{-2}+4z^{-4}$$

This is a polynomial in z^{-1}. No poles except at $z=0$ (from the z^{-4} term). By ROC Property 3 (finite-duration sequences), the ROC is all $z\neq 0$.
Check: $X(1)=2-1+3+0+4=8=\sum_n x[n]$. ✓
Note: the sum of all samples always equals the z-transform evaluated at $z=1$, just as $X(0)$ equals the area under $x(t)$ in the CT Fourier transform. This is a useful sanity check.

19.5 Poles, Zeros, and the ROC

For rational $X(z)$ (a ratio of polynomials in z^{-1}), the ROC is bounded by circles passing through pole locations. The ROC can never contain a pole, because $X(z)\to\infty$ at a pole.

Key Insight

Reading a pole-zero plot:

- Poles ($\times$): values where $X(z)\to\infty$. ROC boundaries are circles through the poles.
- Zeros ($\circ$): values where $X(z)=0$. Zeros do not affect the ROC.
- In the z^{-1} form $X(z)=\frac{1}{1-az^{-1}}$, the pole is at $z=a$ (not at $z=1/a$).
- If $X(z)$ has N poles and $M<N$ finite zeros, there are also $N-M$ zeros at $z=0$.

19.6 Properties of the ROC

ROC Properties for the z-Transform

P1: The ROC is a ring (annulus) centered at the origin: $r_1 < |z| < r_2$.
P2: The ROC does not contain any poles.
P3: **Finite-duration** sequences have ROC = entire z-plane (possibly excluding $z = 0$ or $z = \infty$).
P4: **Right-sided** signals with rational $X(z)$: ROC is exterior, $|z| > \max_i |d_i|$.
P5: **Left-sided** signals with rational $X(z)$: ROC is interior, $|z| < \min_i |d_i|$.
P6: **Two-sided** signals: ROC is an annular ring between consecutive pole circles.
P7: If the ROC includes the unit circle ($|z| = 1$), the DTFT exists.

19.7 The Complete CT/DT Parallel

	Laplace (s-domain)	**z-Transform (z-domain)**
Variable	$s = \sigma + j\omega$	$z = re^{j\omega}$
Transform	$\int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$	$\sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$
FT lives on	$j\omega$-axis ($\sigma = 0$)	unit circle ($\lvert z\rvert = 1$)
ROC shape	vertical strip	annular ring
Right-sided ROC	$\mathrm{Re}\{s\} > \sigma_{\max}$	$\lvert z\rvert > r_{\max}$
Left-sided ROC	$\mathrm{Re}\{s\} < \sigma_{\min}$	$\lvert z\rvert < r_{\min}$
Causal + Stable	all poles in LHP	all poles inside unit circle
Basic pair	$e^{-at}u(t) \leftrightarrow \frac{1}{s+a}$	$a^n u[n] \leftrightarrow \frac{1}{1-az^{-1}}$

19.8 Table of Basic z-Transform Pairs

Signal $x[n]$	$X(z)$	**ROC**
$\delta[n]$	1	all z
$\delta[n-k]$, $k \geq 0$	z^{-k}	$\lvert z\rvert > 0$
$u[n]$	$\dfrac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
$a^n\, u[n]$	$\dfrac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n\, u[-n-1]$	$\dfrac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
$n\, a^n\, u[n]$	$\dfrac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$(n+1)\, a^n\, u[n]$	$\dfrac{1}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$r^n \cos(\omega_0 n)\, u[n]$	$\dfrac{1 - r\cos\omega_0\, z^{-1}}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}$	$\lvert z\rvert > r$
$r^n \sin(\omega_0 n)\, u[n]$	$\dfrac{r\sin\omega_0\, z^{-1}}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}$	$\lvert z\rvert > r$

19.9 Summary

1. The z-transform $X(z) = \sum x[n]\, z^{-n}$ generalizes the DTFT via $z = re^{j\omega}$. Setting $r = 1$ recovers the DTFT.
2. The ROC is essential: without it, $X(z)$ does not uniquely determine $x[n]$.
3. Right-sided $\Rightarrow$ ROC outside the outermost pole. Left-sided $\Rightarrow$ ROC inside the innermost pole. Two-sided $\Rightarrow$ annular ring between pole circles.
4. The **unit circle** ($|z| = 1$) plays the role of the $j\omega$-axis. DTFT exists iff the unit circle is in the ROC.
5. The structure parallels the Laplace transform exactly, with circles replacing lines and $|z|$ replacing $\mathrm{Re}\{s\}$.

19.10 Common Mistakes to Avoid

1. **Omitting the ROC:** $\frac{1}{1-0.5z^{-1}}$ with $|z| > 0.5$ means $(0.5)^n u[n]$, but with $|z| < 0.5$ it means $-(0.5)^n u[-n-1]$.
2. **Confusing "outside" and "inside":** Right-sided signals have ROC *outside* the outermost pole circle. This is different from the Laplace convention ("to the right" of the rightmost pole).

3. **Forgetting that the unit circle replaces the $j\omega$-axis:** The DTFT exists iff the unit circle is in the ROC.
4. **Misidentifying poles:** In $\frac{1}{1-az^{-1}}$, the pole is at $z = a$, *not* at $z = 1/a$.
5. **Geometric series convergence:** $\sum_{n=0}^{\infty} w^n$ converges iff $|w| < 1$, not just $w < 1$. For complex w, the magnitude $|w|$ is what matters.

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